

HDOC DAY – 11

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop. **E.g.** Number: 123, Sum=14
- Find the sum of cubes of its digits using a for loop. **E.g.** Number: 123, Sum= 36
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop. **E.g.** Number: 123, Difference=22

Algorithm:

1. Start and read a positive integer number.
2. If number is not positive, display an error and stop.
3. Display the menu and read choice.
4. Use switch(choice):

Case 1: Set temp = number and sumSquares = 0. While temp > 0,
get digit = temp % 10, add digit*digit to sumSquares,
then set temp = temp / 10. Display sumSquares.

Case 2: Use a for loop with temp = number. In every iteration,
get digit = temp % 10, add digit*digit*digit to sumCubes,
and divide temp by 10. Display sumCubes.

Case 3: Set temp = number. Use a do-while loop to calculate both

sums. Display sumCubes - sumSquares.

Default: Display an invalid-choice message.

5. Stop.

C program:

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int number, choice;
```

```
    int digit, temp;
```

```
    int sumSquares = 0, sumCubes = 0;
```

```
    printf("Enter a positive integer: ");
```

```
    if (scanf("%d", &number) != 1 || number <= 0) {
```

```
        printf("Invalid input. Please enter a positive integer.\n");
```

```
        return 1;
```

```
    }
```

```
    printf("\nMenu\n");
```

```
    printf("1. Sum of squares of digits (while loop)\n");
```

```
    printf("2. Sum of cubes of digits (for loop)\n");
```

```
    printf("3. Difference: sum of cubes - sum of squares (do-while loop)\n");
```

```
    printf("Enter your choice (1-3): ");
```

```
    if (scanf("%d", &choice) != 1) {
```

```
    printf("Invalid menu choice.\n");
    return 1;
}
switch (choice) {
    case 1:
        temp = number;
        while (temp > 0) {
            digit = temp % 10;
            sumSquares += digit * digit;
            temp /= 10;
        }
        printf("Sum of squares of digits = %d\n", sumSquares);
        break;

    case 2:
        for (temp = number; temp > 0; temp /= 10) {
            digit = temp % 10;
            sumCubes += digit * digit * digit;
        }
        printf("Sum of cubes of digits = %d\n", sumCubes);
        break;

    case 3:
        temp = number;
        do {
```

```
    digit = temp % 10;

    sumSquares += digit * digit;

    sumCubes += digit * digit * digit;

    temp /= 10;
} while (temp > 0);

printf("Difference (sum of cubes - sum of squares) = %d\n",
break;

default:

    printf("Invalid choice. Please choose 1, 2, or 3.\n");
}

return 0;
}
```

```

#include <stdio.h>

int main(void)
{
    int number, choice;
    int digit, temp;
    int sumSquares = 0, sumCubes = 0;

    printf("Enter a positive integer: ");
    if (scanf("%d", &number) != 1 || number <= 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 1;
    }

    printf("\nMenu\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference: sum of cubes - sum of squares (do-while loop)\n");
    printf("Enter your choice (1-3): ");

    if (scanf("%d", &choice) != 1) {
        printf("Invalid menu choice.\n");
        return 1;
    }

    switch (choice) {
        case 1:
            temp = number;
            while (temp > 0) {
                digit = temp % 10;
                sumSquares += digit * digit;
                temp /= 10;
            }
            printf("Sum of squares of digits = %d\n", sumSquares);
            break;

        case 2:
            for (temp = number; temp > 0; temp /= 10) {
                digit = temp % 10;
                sumCubes += digit * digit * digit;
            }
            printf("Sum of cubes of digits = %d\n", sumCubes);
            break;

        case 3:
            temp = number;
            do {
                digit = temp % 10;
                sumSquares += digit * digit;
                sumCubes += digit * digit * digit;
                temp /= 10;
            } while (temp > 0);

            printf("Difference (sum of cubes - sum of squares) = %d\n",
                sumCubes - sumSquares);
            break;

        default:
            printf("Invalid choice. Please choose 1, 2, or 3.\n");
    }

    return 0;
}

```

Test case:

number	choice	Expected result
123	1	Sum of squares = $1^2 + 2^2 + 3^2 = 14$
123	2	Sum of cubes = $1^3 + 2^3 + 3^3 = 36$

123	3	Difference = $36 - 14 = 22$
505	1	Sum of squares = $25 + 0 + 25 = 50$
505	2	Sum of cubes = $125 + 0 + 125 = 250$
505	3	Difference = $250 - 50 = 200$
9	3	Difference = $729 - 81 = 648$
0	1	Invalid input message
89	4	Invalid choice message

```
(kali㉿kali)-[~]  
$ ./h11  
Enter a positive integer: 505  
  
Menu  
1. Sum of squares of digits (while loop)  
2. Sum of cubes of digits (for loop)  
3. Difference: sum of cubes - sum of squares (do-while loop)  
Enter your choice (1-3): 3  
Difference (sum of cubes - sum of squares) = 200  
  
(kali㉿kali)-[~]  
$ ./h11  
Enter a positive integer: 9  
  
Menu  
1. Sum of squares of digits (while loop)  
2. Sum of cubes of digits (for loop)  
3. Difference: sum of cubes - sum of squares (do-while loop)  
Enter your choice (1-3): 3  
Difference (sum of cubes - sum of squares) = 648  
  
(kali㉿kali)-[~]  
$ ./h11  
Enter a positive integer: 0  
Invalid input. Please enter a positive integer.  
  
(kali㉿kali)-[~]  
$ ./h11  
Enter a positive integer: 89  
  
Menu  
1. Sum of squares of digits (while loop)  
2. Sum of cubes of digits (for loop)  
3. Difference: sum of cubes - sum of squares (do-while loop)  
Enter your choice (1-3): 4  
Invalid choice. Please choose 1, 2, or 3.
```

```

└─$ ./h11
Enter a positive integer: 123

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 1
Sum of squares of digits = 14

└─(kali@kali)-[~]
└─$ ./h11
Enter a positive integer: 123

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 1
Sum of squares of digits = 14

└─(kali@kali)-[~]
└─$ ./h11
Enter a positive integer: 123

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 2
Sum of cubes of digits = 36

└─(kali@kali)-[~]
└─$ ./h11
Enter a positive integer: 123

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 3
Difference (sum of cubes - sum of squares) = 22

└─(kali@kali)-[~]
└─$ ./h11
Enter a positive integer: 505

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 1
Sum of squares of digits = 50

└─(kali@kali)-[~]
└─$ ./h11
Enter a positive integer: 505

Menu
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference: sum of cubes - sum of squares (do-while loop)
Enter your choice (1-3): 2

```